

# Sprint 5 Intermediate Product Report

Team Name: AI-Driven Robotic Manipulation: Integrating Learning Models with Real-World Control

Team Members: Toan Pham, Albert Vo

Sponsor: Dr. Nakul Gopalan

Date: Nov 18, 2025

## 1. Overview

Sprint Goal: Finalize capstone final report, complete documentation, and prepare comprehensive project presentation and demo

Work Completed:

SBI-501 – Capstone Final Report Completion (In Progress)

## 2. Achievements

User Stories:

Capstone Validation: Final Report and Demo – In Progress

## 3. Risk Management

Foreseen Risks and Challenges: None newly identified this sprint.

Unexpected Risks and Challenges: None reported.

Ongoing Risk Management: Continue tracking schedule and technical risks.

Incorporate Risk Management: Risks are logged and reviewed during sprint meetings.

## 4. Contribution Metrics

### Overall Project Statistics

Total Story Points Completed: 154 (of 170 planned)

Sprint Velocity Average: 31 points

Total GPU Hours: 612

Total Human Hours: ~400 (200 per person)

Lines of Code: 14,326

Documentation Pages: 89

Test Coverage: 68%

## Budget Analysis (Hypothetical)

If using commercial resources:

GPU compute (AWS): ~\$3,500

Software licenses: ~\$500

Human time (RA rate): ~\$8,000

Total value: ~\$12,000

## 5. Evidence

Taiga Board: [https://tree.taiga.io/project/chickeninvader-](https://tree.taiga.io/project/chickeninvader-chickeninvadercross_model_learning_based_robot_control/us/8?kanban-status=10491382)

[chickeninvadercross\\_model\\_learning\\_based\\_robot\\_control/us/8?kanban-status=10491382](https://tree.taiga.io/project/chickeninvader-chickeninvadercross_model_learning_based_robot_control/us/8?kanban-status=10491382)

GitHub Repository: <https://github.com/ChefToan/Cross-Model-Learning-Robot-Control.git>

## 6. Feedback Summary

Feedback was collected during sprint review with the product owner and course staff. Major points were addressed in the backlog.

## 7. Updates

Adjustments: No major changes beyond normal sprint refinement.

New or Withdrawn EPICs/User Stories: Updated on the product backlog as needed.

Changes in Priority: Priorities adjusted based on sprint review outcomes.

## 8. Signature from the Product Owner/Client

Signature/email:

A large, stylized red signature, likely belonging to the Product Owner or Client, written in a cursive, flowing style.